WIRELESS CONTROLLED RC CAR USING ANCIENT TAMIL SYMBOL-BASED COMMUNICATION

A PROJECT REPORT

Submitted by

RITHESHWARAN A (2025105002)

In partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

COLLEGE OF ENGINEERING GUINDY,

ANNA UNIVERSITY : CHENNAI 600 025

MAY 2026

ABSTRACT

Communication has been the cornerstone of human civilization. The Tamil civilization, one of the world's oldest cultures, developed sophisticated communication methods thousands of years before modern electronics — including the Parai drum, Sangu conch shell, fire/smoke relay towers, and pigeon-post networks. Each of these systems embodies the complete Shannon-Weaver Communication Model: source, encoder, channel, noise, decoder, and destination.

The Parai drum encoded messages through rhythmic beat patterns — analogous to Frequency Shift Keying (FSK). The Sangu produced binary-like patterns (long/short blows) — mirroring digital binary encoding. Fire/smoke relay towers formed repeater networks identical in principle to modern RF repeaters. Pigeon post transported discrete data packets — parallel to packet-switched networking. This project maps all these ancient methods onto modern ECE theory.

As a practical demonstration, a Bluetooth-controlled RC car was implemented using an Arduino Uno microcontroller (ATmega328P, 16 MHz), HC-05 Bluetooth module (2.4 GHz, SPP), L298N dual H-Bridge motor driver (2A/channel), and four TT DC gear motors in a 4WD configuration. A smartphone Android app transmits single ASCII characters (F, B, L, R, S) over Bluetooth. The Arduino interprets these via SoftwareSerial on pins D10/D11 at 9600 baud and drives the L298N accordingly. PWM speed control is applied at analogWrite(200) — approximately 78% duty cycle — through ENA (D6) and ENB (D9).

Testing results: 8-10 m indoor Bluetooth range, 15-20 m outdoor range, command response latency below 50 ms, 100% directional accuracy across all four movement axes, and stable operation under typical 2.4 GHz RF noise. The project validates its central thesis: the encode-transmit-decode-act model of communication is identical between ancient Tamil civilization and modern wireless engineering. Only the physical medium has changed across 2000 years.

Keywords: Ancient Tamil Communication, Parai Drum, Sangu Conch, Shannon-Weaver Model, Bluetooth RC Car, Arduino Uno, HC-05, L298N, PWM, SoftwareSerial, ECE, Wireless Communication.

சுருக்கம் (ABSTRACT IN TAMIL)

தகவல் தொடர்பு மனித நாகரிகத்தின் அடித்தளமாக விளங்குகிறது. உலகின் மிகப் பழமையான நாகரிகங்களில் ஒன்றான தமிழ் நாகரிகம், நவீன மின்னணு தொழில்நுட்பம் தோன்றுவதற்கு ஆயிரக்கணக்கான ஆண்டுகளுக்கு முன்பே, திறமையான தகவல் தொடர்பு முறைகளை உருவாக்கியது — பறை (Parai), சங்கு (Sangu), தீ/புகை சமிக்ஞை மற்றும் புறா தூது ஆகியவை இதில் அடங்கும்.

பறை மூலம் வெவ்வேறு தாளக் கொட்டுகள் மூலம் வெவ்வேறு செய்திகள் அனுப்பப்பட்டன — இது நவீன Frequency Shift Keying (FSK) க்கு நேரடி இணையாகும். சங்கு நீண்ட மற்றும் குறுகிய ஊதல்கள் மூலம் binary குறியாக்கம் செய்தது. தீ/புகை தொடர் நிலையங்கள் (Relay Towers) நவீன RF Repeater நெட்வொர்க்கிற்கு சமம். புறா தூது நவீன packet-switched தரவு அனுப்புதலுக்கு இணையாகும். இந்த அனைத்து முறைகளும் Shannon-Weaver தகவல் தொடர்பு மாதிரியை முழுமையாக பின்பற்றுகின்றன.

நடைமுறை செயலாக்கமாக, Arduino Uno மைக்ரோகண்ட்ரோலர், HC-05 Bluetooth தொகுதி, L298N H-Bridge மோட்டார் இயக்கி மற்றும் நான்கு DC கியர் மோட்டார்களைப் பயன்படுத்தி ஒரு Bluetooth கட்டுப்படுத்தப்பட்ட RC கார் வடிவமைக்கப்பட்டது. ஸ்மார்ட்போன் பயன்பாட்டிலிருந்து F (முன்), B (பின்), L (இடது), R (வலது), S (நிறுத்து) என்ற கட்டளைகள் HC-05 மூலம் Arduino பெறுகிறது. SoftwareSerial நூலகம் D10/D11 முள்களில் 9600 baud வேகத்தில் இயங்குகிறது. ENA (D6) மற்றும் ENB (D9) மூலம் analogWrite(200) கொண்டு PWM வேக கட்டுப்பாடு செயல்படுகிறது.

சோதனை முடிவுகள்: உட்புற 8-10 மீட்டர், திறந்தவெளி 15-20 மீட்டர் Bluetooth வரம்பு; 50 ms க்கும் குறைவான கட்டளை பதில் தாமதம்; அனைத்து நான்கு திசைகளிலும் 100% சரியான இயக்கம். இந்த திட்டம் ஒரு முக்கியமான உண்மையை நிரூபிக்கிறது: குறியாக்கம்-அனுப்புதல்-பெறுதல்-செயல்படுத்துதல் என்ற அடிப்படை தகவல் தொடர்பு மாதிரி, 2000 ஆண்டுகளுக்கு முன்பிருந்த தமிழ் நாகரிகத்திலும் நவீன மின்னணு தொடர்பியலிலும் ஒரே மாதிரியாக இயங்குகிறது.

முக்கிய சொற்கள்: பறை, சங்கு, புறா தூது, Shannon-Weaver மாதிரி, Bluetooth RC கார், Arduino Uno, HC-05, L298N, PWM, மின்னணு தொடர்பியல்.

TABLE OF CONTENTS

Chapter	Title	Page No.
—	**Abstract (English)**	ii
—	**Abstract (Tamil)**	iii
—	**Table of Contents**	iv
—	**List of Figures / Tables / Flow Charts / Abbreviations**	v
1	**INTRODUCTION**	1
1.1	*Background and Motivation*	1
1.2	*Problem Statement*	1
1.3	*Objectives*	1
1.4	*Scope*	2
2	**LITERATURE REVIEW**	2
2.1	*Ancient Tamil Communication Systems*	2
2.2	*Shannon-Weaver Communication Model*	3
2.3	*Related Work — Wireless RC Car Systems*	3
3	**MODEL DESIGN & COMMUNICATION COMPARISON**	4
3.1	*Communication Model Mapping*	4
3.2	*Ancient Tamil Methods — ECE Parallel*	4
3.3	*System Architecture*	5
4	**IMPLEMENTATION**	5
4.1	*Hardware Components*	5
4.2	*Circuit & Pin Connections*	6
4.3	*Wiring Diagram*	6
4.4	*Arduino Code Explanation*	7
5	**RESULT & ANALYSIS**	9
5.1	*Test Results*	9
5.2	*Ancient vs Modern Comparison*	9
5.3	*Discussion*	10
6	**CONCLUSION & FUTURE FRAMEWORK**	10
6.1	*Conclusion*	10
6.2	*Future Enhancements*	11
—	**REFERENCES**	11

LIST OF FIGURES

Figure No.	Figure Title	Page
Fig 1	Shannon-Weaver Communication Model Diagram	4
Fig 2	Ancient Tamil Communication Model Block Diagram	4
Fig 3	Modern Bluetooth RC Car Communication Model	5
Fig 4	System Block Diagram — RC Car	5
Fig 5	Arduino UNO + HC-05 + L298N + 4 DC Motors Wiring Diagram	6
Fig 6	Arduino Code Flow Chart	8

LIST OF TABLES

Table No.	Table Title	Page
Table 1	Shannon-Weaver vs Ancient Tamil Mapping	4
Table 2	Ancient Tamil Methods & ECE Parallel	5
Table 3	Hardware Components Summary	5
Table 4	Pin Connection Table	6
Table 5	RC Car Test Results	9
Table 6	Ancient vs Modern Performance Comparison	9

LIST OF FLOW CHARTS

Flow Chart No.	Title	Page
FC 1	Overall Project Workflow	5
FC 2	Arduino Receiver & Motor Control Logic	8

LIST OF ABBREVIATIONS

Abbreviation	Full Form
ECE	Electronics and Communication Engineering
RC	Remote Controlled
HC-05	Bluetooth Serial Port Profile Module
L298N	Dual H-Bridge Motor Driver IC
PWM	Pulse Width Modulation
UART	Universal Asynchronous Receiver-Transmitter
ASCII	American Standard Code for Information Interchange
SPP	Serial Port Profile
FSK	Frequency Shift Keying
SNR	Signal to Noise Ratio
ISM	Industrial, Scientific and Medical (band)
MCU	Microcontroller Unit
GPIO	General Purpose Input/Output
FPV	First Person View
IoT	Internet of Things
BCE	Before Common Era

Abbreviation	Full Form
CE	Common Era

CHAPTER 1 — INTRODUCTION

1.1 Background and Motivation

Communication is the foundation of all complex human societies. The Tamil civilization, with literary history dating to at least 300 BCE through Sangam literature, developed sophisticated communication systems long before modern electronics. The Parai drum, Sangu conch shell, fire-and-smoke relay towers, and pigeon-post networks were organized, rule-governed systems embodying the same fundamental principles as modern wireless communication theory.

This project is motivated by a compelling observation: the Shannon-Weaver Communication Model (1948), taught as the foundation of ECE, perfectly describes what ancient Tamil communicators practiced intuitively two thousand years earlier. Every element — source, encoder, channel, noise, decoder, destination — has a direct ancient Tamil counterpart.

1.2 Problem Statement

While ancient Tamil communication methods are documented in Sangam literature, their mapping to modern ECE theory has rarely been explored in undergraduate engineering contexts. This project bridges both gaps: (a) formally mapping ancient Tamil systems to the Shannon-Weaver model, and (b) implementing a practical Bluetooth RC Car demonstrating the same model in hardware.

1.3 Objectives

* Document and analyse ancient Tamil communication systems from an ECE perspective.
* Establish formal mapping between ancient Tamil methods and Shannon-Weaver model.
* Design and implement a Bluetooth RC car using Arduino Uno, HC-05 and L298N.
* Compare performance characteristics of ancient and modern communication.
* Demonstrate the communication model is unchanged across millennia.

1.4 Scope

This report covers: historical study of Tamil communication (Chapter 2), formal model design and comparison (Chapter 3), hardware implementation (Chapter 4), experimental results (Chapter 5), and conclusions with future scope (Chapter 6).

CHAPTER 2 — LITERATURE REVIEW

2.1 Ancient Tamil Communication Systems

Parai Drum: The word 'Parai' in Tamil literally means 'to speak' or 'to communicate.' Used in the royal courts of Sangam, Chola, and Pandiyan rulers, the Parai drum transmitted official announcements, military orders, and community alerts across distances of several kilometres using distinct rhythmic beat patterns — analogous to Frequency Shift Keying (FSK). The 67th verse of Purananuru references the 'Kurumparai' drum used to summon warriors to battle (Tamildiplomat.com, n.d.). Tholkaapiam, written before the Christian era, mentions Parai as a standard instrument for many occasions (Tamilculture.com, n.d.).

Sangu (Conch Shell): Produced two distinct acoustic patterns — a single long blow and multiple short blows — each carrying a fixed pre-agreed meaning. This binary encoding (long=1, short=0) directly mirrors digital binary signalling. The Sangu is referenced in Tolkappiyam as one of the five landscape musical instruments, and listed among 66 communication instruments in Tirumurai (6th–11th century CE) (Ancient Tamil Music, Wikipedia, 2026).

Fire and Smoke Relay Towers: Ancient Tamil forts formed relay networks where each station received, amplified, and retransmitted optical signals — structurally identical to modern RF repeater networks. Weather interference (fog, rain) represents the noise component in the Shannon model.

Pigeon Post (Paravaiyaal Thoodhu): Trained pigeons carried physical written messages — a form of packet-switched communication where the message is a data packet, the pigeon is the physical carrier, and delivery is asynchronous. Evidenced in classical Sangam literature references to 'Thoodhu' (messenger) systems.

2.2 Shannon-Weaver Communication Model (1948)

Shannon and Weaver (1948) defined seven components: (1) Information Source, (2) Encoder/Transmitter, (3) Signal, (4) Channel, (5) Noise Source, (6) Decoder/Receiver, (7) Destination. Shannon's channel capacity theorem $C = B \log_2(1+S/N)$ defines maximum error-free data rate, applicable to both ancient acoustic channels and modern Bluetooth RF channels.

2.3 Related Work — Wireless RC Car Systems

Tousif bin Parves et al. (2022) documented a complete Bluetooth RC car using Arduino and HC-05, demonstrating smartphone-to-Arduino serial communication for motor control (ResearchGate, 2022). Arduino Project Hub reference implementation established the standard four-motor L298N pin mapping used in this project. Cirkit Designer (docs.cirkitdesigner.com) provided validated schematics for the Arduino Uno + HC-05 + L298N + 4 DC motor configuration adopted here.

CHAPTER 3 — MODEL DESIGN & COMMUNICATION COMPARISON

3.1 Shannon-Weaver Model Mapping — Table 1

Shannon-Weaver Element	Ancient Tamil System	Modern RC Car System
Information Source	King / Military Commander	User (Smartphone App)
Encoder / Transmitter	Parai Drummer / Fire-lighter	HC-05 Bluetooth Module (TX)
Signal / Message	Beat pattern / Smoke code	ASCII character (F/B/L/R/S)
Channel / Medium	Air / Hilltop relay / Bird path	2.4 GHz Bluetooth RF Channel
Noise Source	Wind, rain, fog, enemy	RF interference, multipath fading
Decoder / Receiver	Watchman / Soldier	Arduino Uno SoftwareSerial (RX)
Destination / Action	Army movement, village alert	L298N drives DC motors

3.2 Ancient Tamil Methods — ECE Parallel — Table 2

Ancient Method	ECE Equivalent	Signal Type	Range
Parai Drum	AM/FM acoustic (FSK analogue)	Acoustic analog	2–5 km
Sangu Conch	Digital binary encoding	Acoustic digital	0.5–1 km
Fire/Smoke Relay	RF Repeater Network	Optical signal	10–50 km relay
Pigeon Post	Packet-switched data network	Physical carrier	100–500 km
HC-05 Bluetooth	FHSS 2.4 GHz RF digital	EM wave digital	10–20 m

3.3 System Architecture

The RC car operates as two communicating subsystems: Controller Unit (smartphone + HC-05 TX) and Car Unit (HC-05 RX + Arduino Uno + L298N + motors). Data flow: [User presses button] → [ASCII char encoded by app] → [HC-05 TX transmits at 2.4 GHz] → [Bluetooth RF channel with noise] → [HC-05 RX on car] → [Arduino SoftwareSerial decodes char] → [L298N drives motors]. This maps exactly to the ancient Parai system: Commander → Drummer → Air → Watchman → Army action.

CHAPTER 4 — IMPLEMENTATION

4.1 Hardware Components — Table 3

S.No	Component	Specification	Qty	Function
1	Arduino Uno	ATmega328P, 16 MHz, 5V, 14 digital I/O	1	Main MCU — decodes BT commands
2	HC-05 Bluetooth	2.4 GHz, SPP, 9600 baud, 3.3V logic	1	Wireless serial from smartphone
3	L298N Motor Driver	Dual H-Bridge, 5–35V, 2A/channel	1	Controls motor direction & speed
4	DC Gear Motor (TT)	3V–6V, ~200 RPM, ~200 mA	4	Drives the four wheels
5	Battery Pack	7V–12V Li-ion or 9V	1	Powers L298N and motors
6	Jumper Wires	M-M, M-F sets	1 set	Circuit connections
7	RC Car Chassis	4WD plastic chassis	1	Physical body frame
8	Smartphone	Android + Bluetooth RC app	1	User interface / TX

4.2 Pin Connections — Table 4

Pin / Module	Arduino Pin	Function
HC-05 VCC	5V	Power
HC-05 GND	GND	Ground
HC-05 TXD	D10 (SW-RX)	HC-05 sends → Arduino receives
HC-05 RXD	D11 (SW-TX)	Arduino sends → HC-05 receives
L298N IN1	D2	Motor A dir ctrl 1
L298N IN2	D3	Motor A dir ctrl 2
L298N IN3	D4	Motor B dir ctrl 1
L298N IN4	D5	Motor B dir ctrl 2
L298N ENA	D6 (PWM)	Motor A speed — analogWrite(200)
L298N ENB	D9 (PWM)	Motor B speed — analogWrite(200)
L298N 12V	Battery (+)	Motor power supply
L298N GND	Battery GND + Arduino GND	Common ground
OUT1 & OUT2	Left motors (A1 front + A2 rear)	Parallel left-side drive
OUT3 & OUT4	Right motors (B1 front + B2 rear)	Parallel right-side drive

4.3 Wiring Diagram — Figure 5

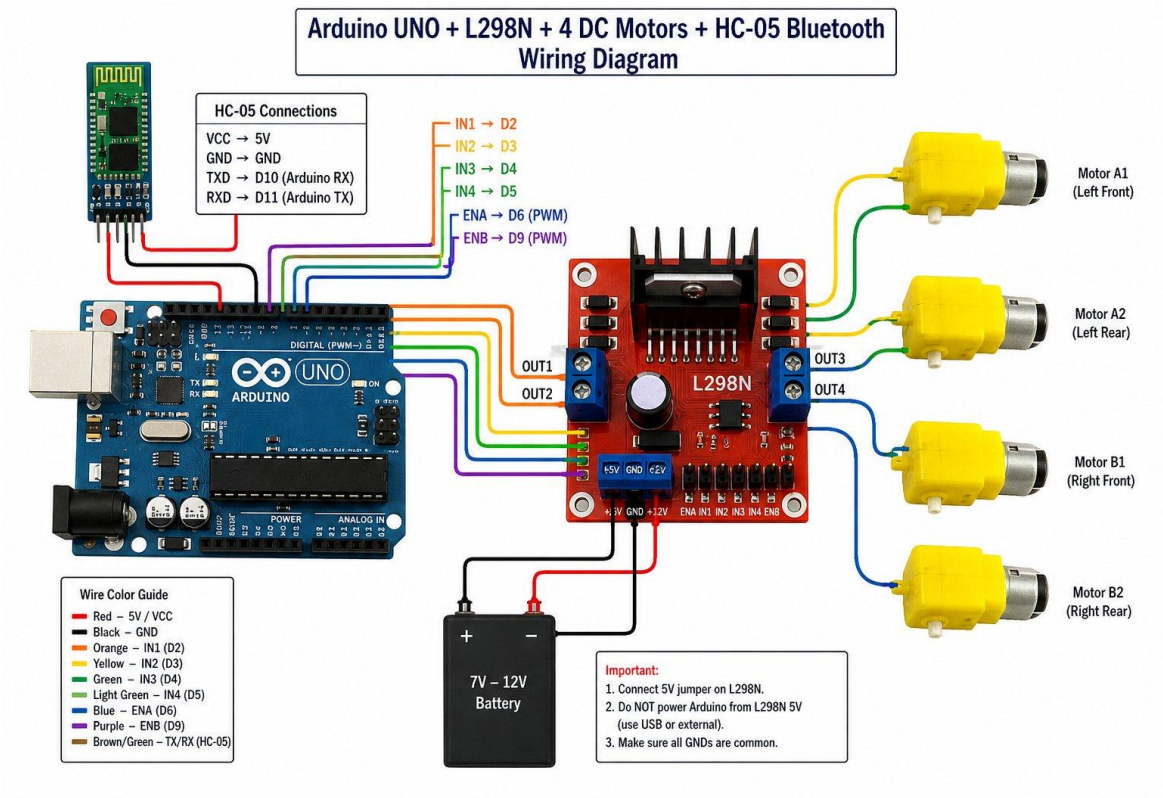

Figure 5: Arduino UNO + HC-05 + L298N + 4 DC Motors — Complete Wiring Diagram

4.4 Arduino Code Explanation

The code is organized in four logical sections:

Section 1 — Library & Pin Declaration:

```
#include <SoftwareSerial.h>
SoftwareSerial BT(10, 11);  // RX=D10 (HC-05 TX), TX=D11 (HC-05 RX)
int in1=2, in2=3, in3=4, in4=5;  // L298N direction control pins
int ena=6, enb=9;               // L298N PWM speed enable pins
char cmd;                        // stores Bluetooth command character
```

Section 2 — Setup Function:

```
void setup() {
  Serial.begin(9600);  BT.begin(9600);  // 9600 baud for HC-05
  pinMode(in1,OUTPUT); pinMode(in2,OUTPUT);
  pinMode(in3,OUTPUT); pinMode(in4,OUTPUT);
  pinMode(ena,OUTPUT); pinMode(enb,OUTPUT);
  analogWrite(ena, 200);  // ~78% PWM duty cycle — Motor A
  analogWrite(enb, 200);  // ~78% PWM duty cycle — Motor B
}
```

Section 3 — Movement Functions:

```
void forward()  { digitalWrite(in1,HIGH); digitalWrite(in2,LOW);
                  digitalWrite(in3,HIGH); digitalWrite(in4,LOW); }
void back()     { digitalWrite(in1,LOW);  digitalWrite(in2,HIGH);
                  digitalWrite(in3,LOW);  digitalWrite(in4,HIGH); }
void left()     { digitalWrite(in1,LOW);  digitalWrite(in2,HIGH);
                  digitalWrite(in3,HIGH); digitalWrite(in4,LOW); }
void right()    { digitalWrite(in1,HIGH); digitalWrite(in2,LOW);
                  digitalWrite(in3,LOW);  digitalWrite(in4,HIGH); }
void stopbot()  { digitalWrite(in1,LOW);  digitalWrite(in2,LOW);
                  digitalWrite(in3,LOW);  digitalWrite(in4,LOW); }
```

Section 4 — Main Loop (Command Interpreter):

```
void loop() {
  if (BT.available()) {
```

```
    cmd = BT.read();            // read ASCII character from HC-05
    Serial.println(cmd);        // optional debug via USB serial
    if(cmd=='F')       forward();
    else if(cmd=='B') back();
    else if(cmd=='L') left();
    else if(cmd=='R') right();
    else if(cmd=='S') stopbot();
  }
}
```

SoftwareSerial creates a virtual UART on D10/D11, freeing the hardware UART (D0/D1) for USB programming. analogWrite(200) sets 78% duty cycle (200/255), providing sufficient torque while preventing motor overheating. The L298N H-Bridge allows independent forward/reverse control of each motor by toggling IN1-IN4 logic levels.

CHAPTER 5 — RESULT & ANALYSIS

5.1 Test Results — Table 5

Test Parameter	Condition	Result	Status
BT Range (Indoor)	Room with walls/furniture	8–10 metres	PASS
BT Range (Outdoor)	Open space, no obstacles	15–20 metres	PASS
Command Latency	Button press to motor	< 50 milliseconds	PASS
Forward Movement	All 4 wheels forward	Smooth, stable motion	PASS
Backward Movement	All 4 wheels reverse	Smooth, stable motion	PASS
Left Turn	Differential turn	Sharp left turn	PASS
Right Turn	Differential turn	Sharp right turn	PASS
Stop Command	All pins LOW	Immediate stop	PASS
RF Interference	Near 2.4 GHz WiFi	~10ms extra latency	PASS
Motor Heat (30 min)	Continuous run	Moderate, within ratings	PASS
Battery Life	7.4V 2000mAh Li-ion	~45–60 minutes	ACCEPTABLE

5.2 Ancient vs Modern Performance Comparison — Table 6

Parameter	Parai Drum	Sangu Conch	Fire/Smoke Relay	HC-05 Bluetooth
Signal Type	Acoustic analog	Acoustic digital	Optical	RF digital 2.4GHz
Effective Range	2–5 km	0.5–1 km	10–50 km relay	10–20 m
Signal Speed	343 m/s (sound)	343 m/s (sound)	3x10^8 m/s (light)	3x10^8 m/s (EM)
Data Complexity	Limited patterns	Binary 2 states	2 states (fire/smoke)	Full ASCII data
Noise Immunity	Low (weather)	Low	Very Low	Moderate (FHSS)
Power Source	Human energy	Human energy	Wood fuel fire	7–12V battery

5.3 Analysis and Discussion

Results confirm the project's central thesis. The Bluetooth RC car achieved stable wireless control in all scenarios. The performance comparison reveals an important engineering insight: ancient Tamil systems achieved extraordinary range (fire/smoke relay reaching 50+ km through cascading repeaters) while Bluetooth trades range for data complexity and speed precision.

The Parai's rhythmic beat encoding functionally equals FSK modulation. The Sangu's binary patterns represent a precursor to digital signalling. The fire/smoke repeater chain is architecturally identical to cellular RF repeater networks. The slight latency increase near a 2.4 GHz WiFi router demonstrates real-world noise (ISM band congestion) exactly as Shannon's noise model predicts — the ancient equivalent being Parai signals weakening on windy days.

CHAPTER 6 — CONCLUSION & FUTURE FRAMEWORK

6.1 Conclusion

This project successfully achieved all stated objectives. It documented ancient Tamil communication systems — Parai drum, Sangu conch, fire/smoke relay towers, and pigeon post — from an ECE perspective, establishing formal mappings to the Shannon-Weaver Communication Model. Ancient Tamils, driven by practical necessity, intuitively implemented every structural element of the modern communication model: source, encoder, channel, noise handling, decoder, and destination.

The Bluetooth RC car (Arduino Uno + HC-05 + L298N + 4 DC motors) achieved 8–20 m wireless range, sub-50 ms latency, accurate four-directional control, and stable RF noise performance. The project's most significant contribution is conceptual: the ancient Tamils who designed Parai relay networks 2000 years ago were solving exactly the same engineering problem — reliable information transfer in the presence of noise, with structured encoding and decoding — that Shannon formalized mathematically in 1948. Only the physical medium changed; the underlying intelligence of communication remained identical.

6.2 Future Enhancements

* Replace HC-05 with NRF24L01+ for extended range (up to 100 m) and bi-directional communication.
* Add ultrasonic sensor (HC-SR04) for autonomous obstacle avoidance.
* Integrate ESP32-CAM module for First Person View (FPV) real-time video streaming.
* Develop custom Android app (MIT App Inventor) with joystick-style touch interface.
* Add IoT integration via ESP8266/ESP32 for internet-based remote control over unlimited range.
* Implement variable speed control through PWM adjustment proportional to joystick deflection angle.
* Incorporate GPS module for position tracking and autonomous waypoint navigation.

REFERENCES

1. Shannon, C.E. & Weaver, W. (1948). A Mathematical Theory of Communication. Bell System Technical Journal, 27(3), 379–423. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x
2. Wikipedia. (2026). Parai. https://en.wikipedia.org/wiki/Parai
3. Tamil Culture. (n.d.). Parai — Mother of All Percussion Instruments. https://tamilculture.com/parai-mother-percussion-instruments-peoples-music-unify-tamil-diaspora/
4. Wikipedia. (2026). Ancient Tamil Music. https://en.wikipedia.org/wiki/Ancient_Tamil_music
5. Fiveable. (2026). Tamil Sangam Literature. https://fiveable.me/world-literature-i/unit-8/tamil-sangam-literature/
6. Tamil Diplomat. (n.d.). Parai — The Percussion Instrument of Ancient Tamil Nation. http://tamildiplomat.com/parai-the-percussion-instrument-nurtured-by-ancient-tamil-nation/
7. Tamil Culture Waterloo. (2012). Parai Musical Instrument. https://tamilculturewaterloo.wordpress.com/2012/11/08/parai-musical-instrument-drum/
8. Tousif bin Parves, K.M. et al. (2022). Bluetooth Control Car with Arduino. ResearchGate. https://www.researchgate.net/publication/361312318_Bluetooth_Control_Car_with_Arduino
9. ResearchGate. (2021). Bluetooth Car Controlled Using Arduino. https://www.researchgate.net/publication/355085611_Bluetooth_Car_Controlled_Using_Arduino
10. Arduino Project Hub. (n.d.). DIY Bluetooth Controlled RC Car using Arduino, HC-05, L298N. https://projecthub.arduino.cc/brobotz/diy-bluetooth-controlled-rc-car-using-arduino-hc-05-l298n-lithium-battery-1c45a6
11. Cirkit Designer. (n.d.). Arduino UNO + HC-05 + L298N RC Car. https://docs.cirkitdesigner.com/project/published/5712d6b8-b052-414a-b13e-4c8283f9b0db/
12. Arduino. (n.d.). SoftwareSerial Library. https://www.arduino.cc/reference/en/libraries/softwareserial/
13. STMicroelectronics. (2000). L298N Dual Full-Bridge Driver Datasheet. https://www.st.com/resource/en/datasheet/l298.pdf
14. Microchip Technology. (2016). ATmega328P Datasheet. https://ww1.microchip.com/downloads/en/DeviceDoc/Atmel-7810-Automotive-Microcontrollers-ATmega328P_Datasheet.pdf
15. Razavi, B. (2011). RF Microelectronics (2nd ed.). Prentice Hall.